Variable-Rate Spraying Shows Promise in Orchards

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Long He, a Penn State ag engineer, speaks during the Fruit Research and Extension Center field day on June 25, 2025, in Biglerville, Pa.

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Heat Wave and Power Outage Can't Stop Adams County Field Day

BIGLERVILLE, Pa. — Digital imaging is poised to reduce chemical thinner usage in orchards.

Long He, a Penn State ag engineering professor, demonstrated the technology June 25 at the university's Fruit Research and Extension Center field day.

Vivid Machines produced the specialized camera, which He mounted on a side-by-side to count the number of fruitlets on each tree.

Using that data, Vivid's software generates a map to prescribe the amount of chemical thinner for a GPS-enabled sprayer to apply to each tree.

"For the trees with more fruits we spray more chemical thinners," said Chenchen Kang, a postdoctoral researcher working with He.

In last year's small trial, the system used 18% less thinner while producing yields and fruit quality similar to those from blocks conventionally sprayed with thinners.

He's team is testing the machine on a larger scale this year, including in a Honeycrisp block at Adams County Nursery.

In that setting the variable-rate system reduced thinner usage by 40% to 50%, and the reductions were especially clear where return bloom was inconsistent, said Shan Kumar, a Penn State tree fruit professor.

The variable-rate technology could be used for other types of spraying beyond chemical thinning.

These applications might allow the technology to provide a net benefit over five to 10 years, Kumar said.

Leasing the camera costs $5,000 per year plus a per-acre cost.

Eventually Vivid Machines and collaborators hope to simplify the two passes into one, with the camera mounted on the sprayer and providing real-time prescriptions.

The two parts of a variable-rate spraying system, a vehicle-mounted camera system and a sprayer that uses the data processed from the camera, are demonstrated June 25, 2025, at the Penn State Fruit Research and Extension Center field day in Biglerville, Pa.

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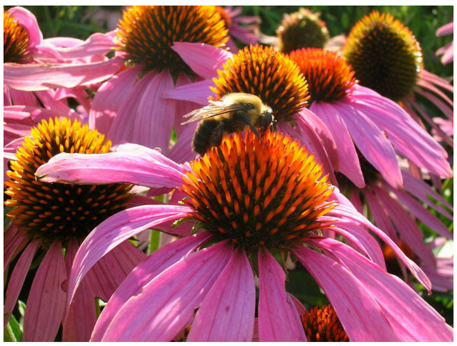
Studying Speedy Bee Helps Cut Its Pesticide Risk



Wild Bees Gain Importance in Orchards

DuBois Farms Thrives Because of Urban Growth

Startup Uses Shrimp Shells To Help Plants Better Absorb Fertilizers and Pesticides

New Tree Fruit Educator Brings Ample Apple Experience

Extreme heat pushed some of the field day sessions indoors, but during the outdoor portion Penn State researchers also demonstrated new technologies they are testing for irrigation and frost protection, as well as techniques for grafting pear trees.

Entomologist David Biddinger said resident bees such as the Japanese orchard bee continue to do yeoman's work to pollinate Adams County fruit trees.

While a honeybee can set 50 apple fruits in a day, a Japanese orchard bee can set more like 2,500.

The introduced species transports pollen on much of its body, while the honeybee only carries pollen on its legs.

Wild bees are increasingly important for orchards. Disease issues have increased the cost of renting honeybee hives, and colonies are sometimes tied up in California almond orchards during Pennsylvania's apple pollination.

But Biddinger doesn't see Japanese orchard bees as a simple replacement for honeybees.

He sees them as a valuable example of the 50 to 60 species of bees that help pollinate orchards in Pennsylvania.

"It's good to have some redundancy in the system, in case one of them goes through a disease cycle or weather conditions are just not right," Biddinger said.

These native bees typically have one generation per year and come out around apricot and red maple bloom.

That means bees need a lot of food early in the season — pollinator strips can help — and their ground nesting sites must be protected from tillage and pesticides.

Pennsylvania has a strong population of pumpkin bees because the state is a leader in no-till pumpkins, Biddinger said.

Resident bees can be far more susceptible to pesticides than honeybees are, but Biddinger said the answer isn't to ban neonicotinoid insecticides. It's to time sprays for windows that reduce the risk to bees.

Spraying for rosy apple aphid at pink exposes bees to 70 to 100 parts per billion of insecticide in the pollen.

Applying insecticide 10 days earlier, at half-inch green, gives the same control of rosy apple aphid but drops bees' exposure below 2 parts per billion, Biddinger said.